

# Practice Test: Calculus Differentiation

1. Derivative of  $f(x) = x^4$ ?

Answer:  $f'(x) = 4x^3$

2. Derivative of  $\cos(x)$ ?

Answer:  $-\sin(x)$

3. Differentiate  $y = \ln(x)$ .

Answer:  $dy/dx = 1/x$

4. Derivative of  $f(x) = 5x^3 - 2x + 1$ ?

Answer:  $f'(x) = 15x^2 - 2$

5. Find the derivative of  $e^{(2x)}$ .

Answer:  $2e^{(2x)}$

6. Differentiate  $y = \tan(x)$ .

Answer:  $dy/dx = \sec^2(x)$

7. Derivative of  $\sqrt{x}$ ?

Answer:  $1 / (2\sqrt{x})$

8. Using the product rule, find the derivative of  $x \sin(x)$ .

Answer:  $\sin(x) + x \cos(x)$

9. Using the quotient rule, find the derivative of  $x / (x+1)$ .

Answer:  $1 / (x+1)^2$

10. Derivative of a constant,  $f(x) = 10$ ?

Answer: 0